Extracting African American English Circuits from GPT-2

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Abstract

Large language models exhibit covert racial bias against speakers of African American English (AAE), but this bias has been characterized almost exclusively at the output level, leaving open where in a model's computation it arises. We construct a contrastive AAE circuit-discovery dataset, and apply Edge Pruning (Bhaskar et al., 2024) to discover and compare circuits for Indirect Object Identification (IOI) and Gendered Pronoun prediction (GP) tasks in GPT-2 Small under matched AAE and Standard American English (SAE) conditions. At 95% edge sparsity, AAE circuits retain only 67–75% of baseline accuracy on IOI variants, while SAE circuits match or exceed baseline. Cross-dialect edge overlap is low (Jaccard 0.25–0.37), yet node-level overlap is substantially higher (0.80) — the two circuits share most of the same components but wire them together differently. Cross-dialect transfer is sharply asymmetric: AAE circuits generalize to SAE inputs (97–110% of same-dialect performance), while SAE circuits fail on AAE inputs (17–50%). We include a preliminary Qwen-2B probe to test whether the directional pattern appears beyond GPT-2. An ablation confirms the divergence is driven by morphosyntactic dialect features, not name identity. To our knowledge, this is the first circuit-level analysis of the divergence of racial dialect processing in LLMs.

1 Introduction

Large language models (LLMs) frequently show covert racial bias against speakers of African American English (AAE): when prompted with AAE inputs, models produce more negative character judgments and harsher decision recommendations than when prompted with Standard American English (SAE) inputs of equivalent meaning (Hofmann et al., 2024). This dialect-based discrimination persists even without explicit racial cues, and existing alignment techniques have been shown to widen rather than close the gap between covert and overt stereotypes (Hofmann et al., 2024; Harvey et al., 2025). As LLMs are increasingly deployed in important sectors such as hiring, education, and criminal justice, understanding the source of this bias has become a significant fairness concern.

To date, however, dialect bias has been characterized almost exclusively at the level of model outputs. Prior work shows that models behave differently on AAE versus SAE inputs (Hofmann et al., 2024; Harvey et al., 2025; Zhou et al., 2025; Mire et al., 2025), but fails to explain where in the model's computation these differences arise. Without a mechanistic understanding of these differences, mitigation efforts are limited to black-box interventions on model outputs, an approach that existing evidence suggests is insufficient and sometimes even counterproductive (Mire et al., 2025; Hofmann et al., 2024).

We address this gap through the lens of *circuits*: sparse computational subgraphs of a model that capture specific aspects of its behavior (Olah et al., 2020). Recent work on automated circuit discovery has produced tools capable of isolating the minimal subset of edges in a Transformer responsible for a given task. Specifically, Bhaskar et al. (2024) introduce Edge Pruning, a gradient-based method that frames circuit discovery as an optimization problem over learnable edge masks. Edge Pruning has been empirically shown to produce sparser and more faithful circuits than prior methods such as ACDC (Conmy et al., 2023), and Edge Attribution Patching (Syed et al., 2023). Compared to earlier automated circuit discovery methods, Edge Pruning additionally scales to settings that previous approaches were unable to reach. These properties make Edge Pruning well-suited for systematic comparisons across multiple tasks and input conditions, helping us study dialect bias mechanistically.

In this work, we apply Edge Pruning to GPT-2 Small to investigate whether AAE and SAE inputs are processed by structurally different circuits within the same model, and include a preliminary probe on the larger Qwen-2B. We construct a contrastive AAE dataset by applying Multi-VALUE (Ziems et al., 2023) dialect transformations to standard circuit-discovery benchmarks, including Indirect Object Identification (IOI-t1 and IOI) (Wang et al., 2022) and Gendered Pronoun prediction (Mathwin et al., 2023), from the original (SAE) dataset, creating matched sentence pairs that differ only in dialect-specific morphosyntactic features. We run Edge Pruning separately for each (dialect, task) pair and compare the resulting circuits along three axes: task accuracy and faithfulness, edge-level overlap, and cross-dialect transfer. Together, these comparisons let us ask whether dialect-specific circuits exist and whether they constitute distinct computational pathways or only variations of a shared mechanism.

In this paper, we make the following contributions:

- We construct a contrastive AAE dataset for circuit discovery using Multi-VALUE (Ziems et al., 2023) rule-based dialectal translation over IOI and GP.
- We discover and characterize dialect-specific circuits in GPT-2 Small, finding that AAE circuits are consistently less faithful and lower-accuracy than their SAE counterparts at matched sparsities.
- We show that within-dialect circuit similarity exceeds cross-dialect similarity, and that cross-dialect transfer is asymmetric: AAE circuits generalize to SAE inputs, while SAE circuits fail on AAE inputs. This constitutes, to our knowledge, the first circuit-level evidence that dialect bias in a language model has structural character.
- We construct new code extending the Edge Pruning framework (Bhaskar et al., 2024) to support (i) a Multi-VALUE-based AAE dataset construction pipeline with single-token name filtering, task-preserving rule disabling, and post-translation pronoun restoration; (ii) cross-dialect transfer and edge/node-level circuit overlap evaluation harnesses absent from the original codebase; and (iii) a port of the pipeline to Qwen-2B, including tokenizer-adapted name filtering and architecture-specific configuration for circuit discovery on a model without native task capability.

2 Related Work and Background

2.1 Dialect bias in language models.

LLMs exhibit measurable performance disparities across dialect varieties of English. Prior work has shown that LLMs associate AAE speakers with negative character traits and recommend harsher outcomes based on dialect cues (Hofmann et al., 2024; Fleisig et al., 2024; Groenwold et al., 2020). This covert bias is stronger than overt bias: when race is made explicit, models temper their responses, but dialect-based discrimination persists and intensifies. A recent quality-of-service framing introduces a framework for auditing LLM-based chatbots for dialect bias, with a case study demonstrating that a deployed chatbot produces vaguer or less accurate responses to AAE prompts than to matched SAE prompts (Harvey et al., 2025). Zhou et al. (2025) further document that LLMs produce less accurate responses and simpler reasoning chains and explanations for AAE inputs than for matched SAE inputs. Mire et al. (2025) demonstrate that reward models systematically penalize AAE-dialect responses, suggesting that Reinforcement Learning from Human Feedback (RLHF)-based alignment may reinforce rather than correct dialect-based discrimination. Across these works, the consistent finding is that dialect bias is not incidental noise but a systematic property of current LLMs. What remains unaddressed is the internal mechanism driving it.

2.2 Mechanistic interpretability and circuits.

Mechanistic interpretability aims to decompose neural network behavior into human-understandable components (Olah et al., 2020; Elhage et al., 2021). The central object of study is a *circuit*: a sparse computational subgraph of a model's full computational graph that is causally responsible for a particular behavior. Circuits were originally identified through manual analysis: Wang et al. (2022) hand-identified the circuit responsible for Indirect Object Identification in GPT-2 Small; Mathwin et al. (2023) did the same for gendered pronoun predictions. ACDC (Conmy et al., 2023) automated circuit discovery via greedy activation patching, which iteratively ablated each edge and retained those whose removal degraded task performance. While a significant advance, ACDC is computationally prohibitive: its runtime scales linearly with the number of edges, making it impractical beyond a few hundred examples or billion-parameter models. Edge Attribution Patching (EAP; Syed et al., 2023) achieves better scalability by approximating edge importance via first-order gradient attribution, assigning each edge a score in a single forward-backward pass. However, this linear approximation ignores interactions between edges and consistently produces suboptimal circuits.

2.3 Edge Pruning.

Bhaskar et al. (2024) reframe circuit discovery as an optimization problem and propose Edge Pruning as its solution. Rather than greedy search or first-order approximation, Edge Pruning introduces continuous, learnable binary masks over the edges of a Transformer's computational graph and optimizes them via gradient descent with L_0 regularization (Louizos et al., 2018). Missing edges are replaced with counterfactual activations from corrupted examples under interchange ablation (Geiger et al., 2020), keeping the residual stream in-distribution. To support edge-level masking, Edge Pruning replaces the standard residual stream with a *disentangled residual stream* (Friedman et al., 2023) that retains all prior layer activations, allowing each node to receive a dynamically-weighted combination of clean and corrupted inputs. At convergence, continuous masks are thresholded to binary values, yielding the final circuit. On four standard circuit-finding tasks in GPT-2 Small, Edge Pruning recovers circuits that are more faithful and sparser than those found by ACDC or EAP, achieving $2.65\times$ fewer edges than ACDC at equivalent faithfulness on multi-template IOI. It scales to 100K training examples without degrading, and recovers ground-truth circuits exactly in two Tracr-compiled models (Lindner et al., 2023). At the extreme end, Edge Pruning scales to CodeLlama-13B, identifying circuits with just 0.04% of model edges that match the full model's performance on Boolean Expressions.

2.4 Component analysis and bias.

Prior research also extends lower level analysis to socially-relevant model behavior, primarily through the exploration of bias. Prakash and Roy (2024) introduce a slightly higher-level feature-based approach to localize gender bias in LLaMA-2-7B, LLaMA-3-8B, and Mistral-7B. They do so by combining activation and attribution patching to trace how bias originates in early MLPs and how it propagates through middle-layer attention heads to the final token. While their analysis identifies components relevant to gender bias, they stop a step above circuit-level analysis since connections between components are inferred rather than found through edge-level interventions. Chandna et al. (2025) take this a step further and identify edge-level circuits for demographic and gender bias in their work with GPT-2 Small, GPT-2 Large, and LLaMA-2 (Chandna et al., 2025). Their findings conclude that bias-related circuits are disjoint across bias categories, meaning the underlying representations are modular and specialized. While targeted edge ablations can attenuate biased output, there were also effects on other tasks, indicating some level of entanglement between bias circuits and general linguistic competence in models.

These studies establish that demographic biases in model outputs have identifiable mechanistic signatures and can be localized at the component of edge level. However, they also largely focus on gender and demographic bias in SAE, leaving a gap for how models process linguistic differences across dialects. Our work addresses this gap by characterizing the mechanistic divergence between AAE and SAE, providing a foundation for future research on dialect-related bias.

3 Methodology

3.1 Tasks

We study three tasks from the mechanistic interpretability literature. Indirect Object Identification (IOI) requires the model to predict the indirect object name given a sentence of the form "When Mary and John went to the store, John gave a bottle of milk to ___" (Wang et al., 2022). We study two variants: IOI (multi-template) and IOI-t1 (single template). IOI-t1 serves as a simpler, more controlled condition with less template-induced variance (Bhaskar et al., 2024). Gendered Pronoun prediction (GP) requires predicting a gendered pronoun given an occupational context, e.g., "The nurse said that ___" (Mathwin et al., 2023). Both tasks have well-characterized circuits in GPT-2 Small and involve syntactic constructions susceptible to AAE morphosyntactic variation. The Greater Than task is omitted as it offers limited contrastive signal across dialects.

3.2 Dataset Construction

Datasets follow a 2×2 design crossing dialect (SAE, AAE) with task (IOI, GP).[1] SAE datasets are taken directly from (Bhaskar et al., 2024), with splits mirroring the original paper: IOI-t1 at 50/50/50, IOI at 200/200/36, and GP at 150/150/378 (train/val/test).

AAE datasets are produced by applying Multi-VALUE (Ziems et al., 2023) rule-based dialect translation to each SAE sentence. Rule-based translation was preferred over LLM-based approaches such as AAVENUE (Gupta et al., 2025) for its determinism, linguistic validation, and lower risk of hallucinations that could corrupt task-relevant syntax. Two rules were disabled to preserve task validity: `clefting` and `double_obj_order` for IOI, both of which would invalidate the final-token prediction target; for GP, pronouns neutralized by Multi-VALUE were restored post-translation. 4.0% of IOI examples exhibiting residual final-token drift were excluded, introducing a mild selection bias toward morphologically simpler templates.

SAE names were replaced with African American names drawn from a list of the top 1,000 African American baby names (Successful Black Parenting, 2024), filtered to single tokens under the GPT-2 BPE vocabulary, yielding a pool of 487 names. For Qwen-2B, an independent filter left 448 of 487 names; examples with multi-token predicted names were additionally dropped, retaining approximately 92% of examples per split.

3.3 Circuit Discovery

Edge Pruning (Bhaskar et al., 2024) is applied separately to each of the six (dialect × task) conditions per model. Following the original paper, we freeze model weights and optimize continuous edge masks via gradient descent with L_0 regularization, replacing removed edges with counterfactual activations from corrupted examples under interchange ablation. Hyperparameters mirror those of Bhaskar et al. (2024): edge sparsity target 0.95, node sparsity 0.72 (IOI) and 0.69 (GP), learning rate 0.8, 3,000 optimization steps with 2,500 warmup steps, and `-disable_node_loss`. At convergence, continuous masks are thresholded to binary values to yield the final circuit.

3.4 Evaluation Metrics

Circuits are evaluated along four axes:

- Task performance: Accuracy, KL divergence, and logit difference—the log-probability gap between correct and incorrect target tokens—at 95% edge sparsity relative to the unpruned baseline.
- Faithfulness: Exact match and KL divergence between circuit and full-model output logits on the held-out test split.
- Circuit overlap: Jaccard similarity $|A \cap B|/|A \cup B|$ between edge sets, computed for cross-dialect pairs (AAE vs. SAE, same task) and within-dialect cross-task pairs (IOI vs. IOI-t1), with the latter serving as a baseline.
- Cross-dialect transfer: Accuracy evaluating each circuit on the opposite dialect's test split, as a percentage of same-dialect performance to quantify asymmetry.

4 Results

4.1 Circuit Faithfulness and Accuracy

Table 1 reports circuit accuracy, exact match (EM), and KL divergence for all six GPT-2 Small circuits at 95% edge sparsity.[2] The *ratio* column expresses circuit accuracy as a fraction of unpruned baseline accuracy. SAE circuits match or exceed baseline accuracy in every case (ratios 1.00–1.17). AAE circuits fall below baseline in every case where the baseline is above floor: AAE-IOI retains 74.8% of baseline accuracy (0.435 vs. 0.582), and AAE-IOI-t1 retains 66.7% (0.327 vs. 0.490). The AAE-GP circuit matches its baseline exactly (ratio 1.00), but this reflects faithful reproduction of an already-weak model—the circuit reproduces a 0.362 baseline with high precision (EM,=,0.815, KL,=,0.198). AAE circuits also show substantially higher KL divergence than their SAE counterparts, indicating greater distributional divergence from the full model even when point accuracy is comparable.

[1] We additionally construct IOI-t1 as discussed in Section 3.1

[2] SAE conditions reproduce the faithfulness–sparsity trends reported by Bhaskar et al. (2024), validating our experimental pipeline.

Circuit	acc_{base}	$acc_{circuit}$	Ratio	EM	KL
SAE-IOI	0.795	0.795	1.00	0.822	0.210
AAE-IOI	0.582	0.435	0.75	0.556	0.661
SAE-t1	0.860	0.980	1.14	0.980	0.116
AAE-t1	0.490	0.327	0.67	0.510	0.656
SAE-GP	0.587	0.688	1.17	0.937	0.059
AAE-GP	0.362	0.362	1.00	0.815	0.198

Table 1: GPT-2 Small circuit faithfulness at 95% edge sparsity. *Base acc.* is unpruned model accuracy; *Ratio* is circuit accuracy divided by base accuracy. SAE circuits match or exceed baseline in every condition; AAE circuits fall below baseline for IOI and IOI-t1, and match a low baseline for GP.

4.2 Circuit Overlap

Table 2 reports Jaccard similarity and circuit sizes for cross-dialect and within-dialect circuit pairs. Cross-dialect Jaccard for GPT-2 Small ranges from 0.247 (IOI-t1) to 0.367 (IOI), comparable to the within-dialect cross-task baselines (AAE: 0.382, SAE: 0.260), indicating that same-task circuits across dialects are no more structurally similar than different-task circuits within a dialect. Figure 1 shows the layer-wise distribution of shared, SAE-only, and AAE-only edges for IOI and GP. In the IOI panel, AAE-only edges outnumber SAE-only edges in every layer from L0 to L8, while SAE-only edges outnumber AAE-only edges in L9–L11.

Head-type analysis. Wang et al. (2022) identified seven functional head types in the GPT-2 Small IOI circuit: name movers, backup name movers, negative name movers, S-inhibition heads, induction heads, duplicate token heads, and previous token heads. All 20 Wang-taxonomy heads recovered by Edge Pruning in the SAE-IOI circuit are retained in the AAE-IOI circuit, with zero Wang-classified heads exclusive to SAE. The AAE circuit additionally recruits two supplementary functional heads absent from SAE: L0H10 (duplicate token) and L5H8 (induction). The all-head Jaccard is 0.802, substantially higher than the edge-level Jaccard of 0.367, indicating that the two circuits share most of the same nodes but differ substantially in their connectivity. Figure 2 displays this structure.

4.3 Cross-Dialect Transfer

Table 3 reports exact match when each circuit is evaluated on the opposite dialect's test split. The results reveal a consistent and strongly asymmetric transfer pattern across all tasks. For GPT-2 Small, AAE circuits transfer to SAE inputs with minimal degradation: the AAE-IOI circuit achieves EM,=,0.803 on SAE inputs, retaining 97.7% of its same-dialect performance. Transfer in the reverse direction is markedly worse: the SAE-IOI circuit achieves only EM,=,0.407 on AAE inputs (49.5% of same-dialect performance), and SAE-IOI-t1 drops to 16.7%. To determine whether this generalizes beyond GPT-2, we evaluate on Qwen-2B (Appendix C), where the asymmetry replicates directionally despite that model lacking native IOI capability.

Task	$\|A\|$	$\|B\|$	$\|A \cap B\|$	Jacc.	OC
GPT-2: cross-dialect (AAE vs. SAE)					
IOI	1625	1417	816	0.367	0.576
GP	1260	1039	501	0.279	0.482
IOI-t1	1404	1192	513	0.247	0.430
GPT-2: within-dialect cross-task (IOI-t1 vs. IOI)					
AAE	1404	1625	837	0.382	0.596
SAE	1192	1417	539	0.260	0.452
Qwen-2B: cross-dialect (AAE vs. SAE)					
IOI	12925	10999	4901	0.258	0.446
GP	10421	11579	2626	0.136	0.252

Table 2: Circuit overlap statistics. Within each section, $|A|$ and $|B|$ are the edge counts of the two circuits being compared, in the order listed in the section header. OC is overlap coefficient ($|A \cap B|/\min(|A|, |B|)$). Cross-dialect Jaccard for GPT-2 Small (0.247–0.367) is comparable to within-dialect cross-task Jaccard (0.260–0.382), indicating that same-task circuits across dialects are no more structurally similar than different-task circuits within a dialect.

	IOI		IOI-t1		GP	
Circuit	SAE	AAE	SAE	AAE	SAE	AAE
GPT-2 Small						
SAE	**0.822**	0.407	**0.980**	0.163	**0.937**	0.415
AAE	0.803	**0.556**	0.960	**0.510**	0.894	**0.815**
Qwen-2B (IOI only)						
SAE	**0.734**	0.215	—	—	—	—
AAE	0.730	**0.288**	—	—	—	—

Table 3: Cross-dialect transfer: exact match (EM) for each circuit evaluated on SAE and AAE test splits. Bold values are same-dialect (diagonal) performance. AAE circuits transfer to SAE with minimal loss; SAE circuits fail substantially on AAE inputs across all tasks and both models.

4.4 Ablation: SAE Syntax with AAE Names

To test whether the circuit divergence is driven by name identity rather than morphosyntactic dialect

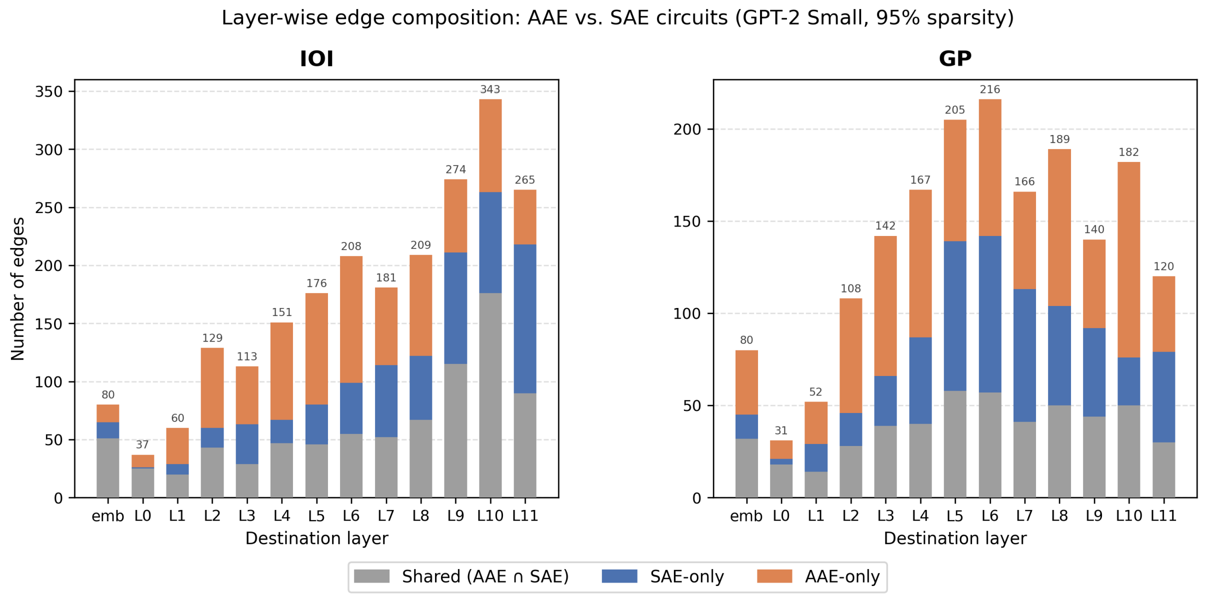

Figure 1: Layer-wise edge composition for AAE and SAE circuits in GPT-2 Small at 95% sparsity, for IOI (left) and GP (right). Each bar shows the number of edges with a given destination layer, split into shared edges (grey), SAE-only edges (blue), and AAE-only edges (orange). In IOI, AAE-only edges are concentrated in early-to-middle layers (L2–L8), while SAE-only edges are heavier in late layers (L9–L11).

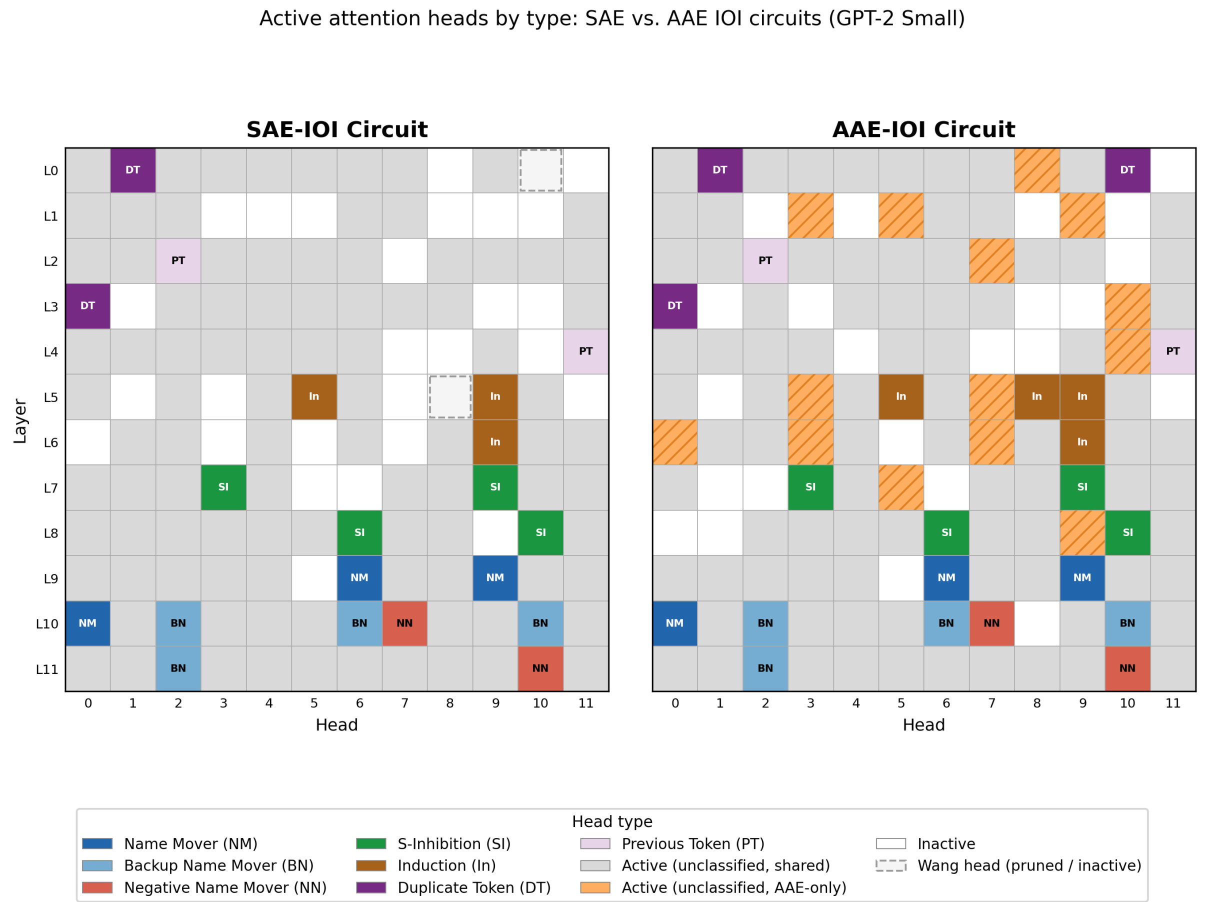

Figure 2: Active attention heads by type for SAE-IOI (left) and AAE-IOI (right) circuits in GPT-2 Small. Cells are colored by Wang et al. (2022) head type where applicable; grey cells are active but unclassified; orange hatched cells are active in the AAE circuit only; dashed-border cells are Wang-classified heads inactive in the respective circuit. All canonical IOI heads are shared; circuit divergence is concentrated in unclassified heads.

features, we trained circuits on SAE sentences in which the original names were replaced with the AAE name set used in our primary experiments (the *ablation* condition). These sentences preserve SAE syntax while using African American names, isolating the contribution of lexical identity from dialectal structure.

The ablation circuits do not bridge the gap toward AAE. When evaluated on AAE test inputs, the ablation-IOI circuit achieves EM = 0.206 and Jaccard = 0.306 with the AAE-IOI circuit—lower than the standard SAE circuit's EM = 0.407 and Jaccard = 0.367 on the same inputs. Results are consistent across tasks: ablation IOI-t1 achieves EM = 0.122 on AAE-IOI-t1 inputs (vs. SAE EM = 0.163), and ablation GP achieves EM = 0.606 on AAE-GP inputs (vs. SAE EM = 0.415). Conversely, the AAE circuit transfers to ablation inputs nearly as well as to standard SAE inputs (IOI: EM = 0.768 vs. 0.803; GP: EM = 0.852 vs. 0.894), confirming that the AAE circuit's SAE generalizability does not depend on the specific names present. Circuit divergence is therefore attributable to morphosyntactic features, not name identities.

5 Discussion

Our results organize around two main findings.

1. AAE circuits are less faithful and less compressible than SAE circuits. Across all three tasks, SAE circuits match or exceed unpruned baseline accuracy, while AAE circuits retain only 67–75% of their already-lower baselines on IOI variants and faithfully reproduce a weak baseline on GP. We attribute this to AAE being weakly represented in a model trained overwhelmingly on SAE-dominant text: Edge Pruning isolates a sparse, well-behaved subgraph for SAE because the relevant computation is concentrated, whereas AAE computation appears more distributed and harder to compress without accuracy loss.

2. The AAE circuit is a superset of the SAE core. Cross-dialect transfer is sharply asymmetric: AAE circuits process SAE inputs with minimal loss (97–110% of same-dialect EM), while SAE circuits fail on AAE inputs (17–50%). The AAE circuit contains the SAE core plus additional heads for AAE-specific morphosyntax — specifically, supplementary duplicate-token and induction heads absent from the SAE circuit — and SAE circuits, lacking these, cannot route AAE inputs correctly. The dissociation between edge-level Jaccard (0.367) and node-level Jaccard (0.802) confirms that the circuits share most of the same heads but wire them together differently. Our ablation localizes the divergence to morphosyntactic structure rather than name-based associations: pairing SAE syntax with AAE names moves circuits *further* from the AAE circuit (Jaccard 0.306) than the standard SAE circuit is (0.367).

These findings have a direct fairness implication: fixing a model's SAE behavior provides no guarantee of fixing its AAE behavior, since SAE circuits lack the additional heads required to process AAE inputs. Effective mechanistic interventions will require an understanding of these specialized pathways.

We also note that although Qwen-2B lacks native IOI capability, it replicates the directional pattern: larger AAE circuits, lower cross-dialect Jaccard, and the same transfer asymmetry. While systematic cross-model replication on task-capable models is desirable, it was unfortunately out of our computational scope, and we leave it to future work.

6 Limitations

Several methodological constraints limit the conclusions we can draw. First, our AAE name pool is restricted to names that tokenize as a single token under the GPT-2 BPE vocabulary. This constraint systematically excludes many distinctively African American names — Lakeisha, Shaniqua, Aaliyah, and similar names that are phonologically distant from the SAE-dominant vocabulary — biasing the AAE condition. The direction of this bias is conservative: if anything, it understates the dialect contrast we are trying to measure. Second, Multi-VALUE's rule-based translation applies morphosyntactic transformations but does not capture prosodic, discourse-level, or lexical features of AAE. Our AAE datasets represent a narrow, morphosyntax-only dialect contrast; results may differ with richer dialect representations. Third, 4% of AAE-IOI training and validation examples were excluded due to final-token drift introduced by Multi-VALUE's possessive suffix and gerund constructions on proper nouns. While the rate is low and consistent across splits, this introduces a mild selection bias toward morphologically simpler templates. Additionally, Multi-VALUE's pronoun neutralization replaced gendered prediction targets with neutral forms; we had to restore the original pronouns post-translation to maintain the objective, but this might have additionally deviated from natural AAE text. Our Qwen results are from a base model without native IOI capability; to achieve a true comparison to GPT-2 would require task-specific fine-tuning. Finally, all of our contrastive AAE circuit discovery runs use a fixed 95% edge sparsity target; the faithfulness–sparsity relationship may differ between AAE and SAE circuits at other operating points, which we do not fully characterize.

7 Conclusion and Future Work

We applied Edge Pruning to GPT-2 Small to discover and compare circuits for Indirect Object Identification and Gendered Pronoun prediction under matched AAE and SAE conditions. Our results support two main conclusions: AAE circuits are less faithful and less compressible than their SAE counterparts, consistent with AAE computation being more distributed in a model trained on SAE-dominant text; and the AAE circuit is an approximate superset of the SAE core, containing additional duplicate-token and induction heads used for processing AAE-specific morphosyntax, which explains the sharp asymmetry in cross-dialect transfer. Together, these constitute the first circuit-level evidence that LLMs do not merely produce different outputs for AAE inputs, but process them through a meaningfully different internal mechanism.

Future work should extend this analysis to large frontier models. It should additionally move beyond syntactic proxy tasks to mechanistically analyze semantically biased outputs — such as negative character judgments toward AAE speakers — to understand bias beyond LLM understanding, and better address real-world tasks where the societal stakes are highest.

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A Generative AI Disclosure

Generative AI was used to support this project at all stages, including planning evaluation structure, writing code, and editing prose.

B Unpruned Baselines

Table 4 reports unpruned GPT-2 Small accuracy across all six dialect–task conditions. The model exhibits a substantial accuracy gap between dialects prior to any circuit discovery: SAE accuracy exceeds AAE accuracy by 21.3 percentage points on IOI, 37.0 points on IOI-t1, and 22.5 points on GP. This baseline disparity is consistent with prior work on dialect-based quality of service disparities (Harvey et al., 2025) and contextualizes the circuit-level comparisons that follow.

Task	SAE acc.	AAE acc.	Gap
IOI	0.795	0.582	−0.213
IOI-t1	0.860	0.490	−0.370
GP	0.587	0.362	−0.225

Table 4: Unpruned GPT-2 Small accuracy on SAE and AAE test splits. The model shows a consistent accuracy deficit on AAE inputs across all tasks before any circuit manipulation.

C Qwen-2B: Accuracy and Faithfulness

The Qwen-2B base model does not exhibit IOI or GP behavior: baseline logit difference is negative for both tasks (IOI: -5.29; GP: -2.10), and baseline exact match is near zero (IOI: 0.007; GP: 0.024). Edge Pruning applied to Qwen therefore does not recover a pre-existing task circuit; rather, it identifies the sparse subgraph that best approximates the IOI logit signal under interchange ablation in a model without native task capability. We report these results as a preliminary cross-architecture probe, with the caveat that they are not directly comparable to the GPT-2 results.

Table 5 reports IOI circuit faithfulness for Qwen-2B. Despite the near-zero baseline, Edge Pruning recovers a subgraph with meaningful IOI accuracy for the SAE condition (0.749 circuit accuracy, EM = 0.734). The AAE circuit achieves substantially lower accuracy on its own dialect (0.115, EM = 0.288), mirroring the direction of the GPT-2 faithfulness gap.

For GP, Qwen circuit accuracy is at floor for both dialects ($\leq$0.005); exact match values are 0.471 (SAE) and 0.302 (AAE), preserving the directional pattern but providing insufficient signal for strong conclusions. We report Qwen GP results for completeness but do not draw architectural generalisations from them.

Circuit	Base acc.	Circ. acc.	EM	KL
SAE-IOI	0.001	0.749	0.734	0.431
AAE-IOI	0.000	0.115	0.288	0.658

Table 5: Qwen-2B IOI circuit faithfulness at 95% edge sparsity. Base model accuracy is near zero, reflecting the absence of native IOI behaviour; circuits reflect latent structure recoverable by sparse subgraph isolation rather than an established task circuit.